

Face Mask Detection System

Design Document

Version 1.0



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Revision History

Date (dd/mm/yyyy)	Version	Description	Author
20-01-2026	1.0	This project aims to develop an automated face mask detection system using deep learning techniques. The system will identify whether a person in an image is wearing a mask or not wearing a mask. A Convolutional Neural Network (CNN) model will be trained on a dataset containing face images with and without masks. The model will be integrated with a real-time detection interface to monitor live camera feeds or uploaded images. This technology can be applied in public safety monitoring, hospital check-ins, and smart surveillance systems	BC220404931

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1. Introduction of Design Document

This design document provides a detailed overview of our software project. It explains how the system is structured, how different parts of the system interact, and how data flows through it. The document includes diagrams, database design, user interfaces, and test cases to give a complete understanding of the system before development.

The purpose of this document is to serve as a guide for developers, testers, and stakeholders. It helps ensure that everyone understands the project requirements, system functionality, and design decisions. By preparing this document, we can reduce errors, improve communication, and make the development process more organized and efficient.

This Design Document presents the detailed design specifications for the **Face Mask Detection System**. It serves as a bridge between the requirements gathered in the earlier phases and the actual implementation of the system.

Purpose of this Document:

The primary purpose of this document is to provide a comprehensive blueprint for developers, testers, and stakeholders to understand how the system will be built. It translates the functional and non-functional requirements into technical designs that guide the development process.

What is Included in this Document:

This document contains the following key design components:

- **Entity Relationship Diagram (ERD)** – Shows the database structure, including all entities, their attributes, and relationships between them.
- **Sequence Diagrams** – Illustrate the step-by-step interactions between system components for each use case.
- **Architecture Design** – Presents the overall system architecture, showing how different layers and components interact.
- **Class Diagram** – Defines the object-oriented structure of the system, including classes, attributes, methods, and their relationships.
- **Database Design** – Provides detailed database schema with tables, columns, data types, and constraints.
- **Interface Design** – Shows visual mockups of the main user interfaces to demonstrate the look and feel of the application.

- **Test Cases** – Contains test scenarios to verify that each system function works correctly.

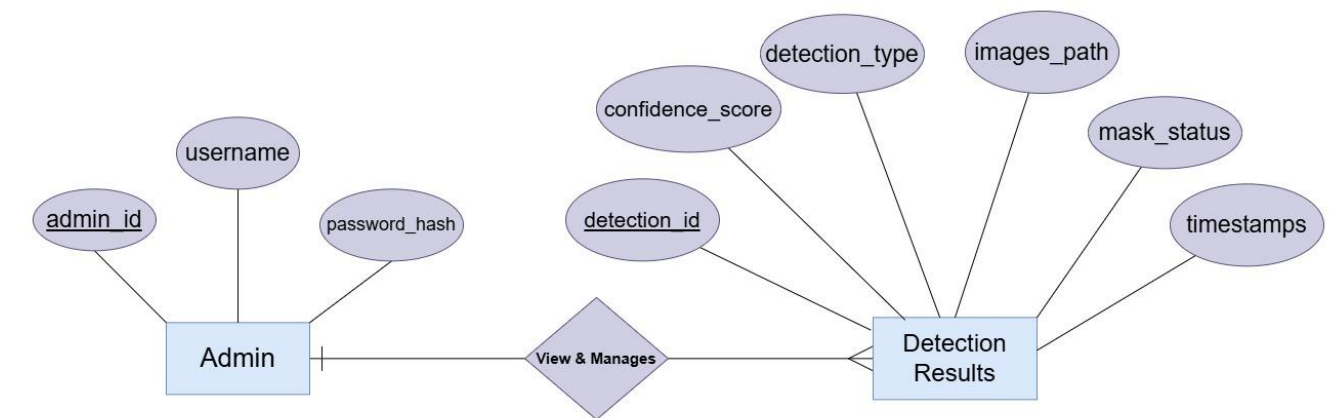
Benefits of the Design Phase:

The design phase is crucial in software development for several reasons:

1. **Reduces Development Risks** – Identifying design issues early prevents costly changes during implementation.
2. **Improves Communication** – Provides a common understanding among team members, stakeholders, and developers.
3. **Ensures Quality** – Well-planned design leads to a more maintainable, scalable, and robust system.
4. **Guides Implementation** – Serves as a roadmap for developers, reducing confusion and rework.
5. **Facilitates Testing** – Test cases can be prepared in advance based on the design specifications.

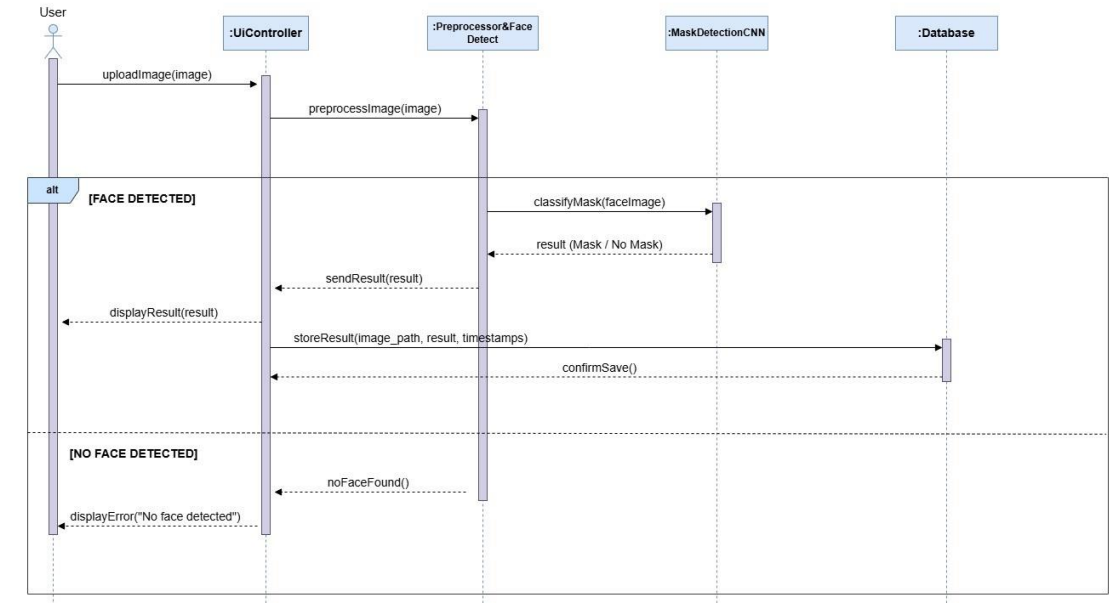
This document will be used as a reference throughout the development and testing phases to ensure that the final system meets all specified requirements.

2. Entity Relationship Diagram (ERD)

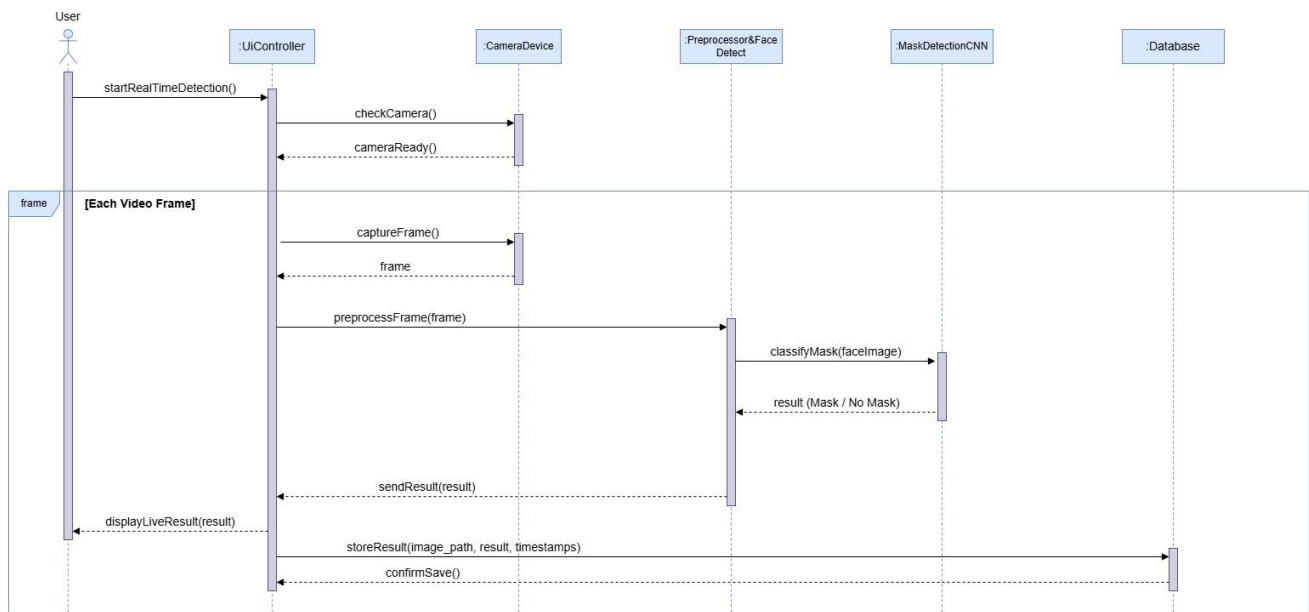


3. Sequence Diagrams

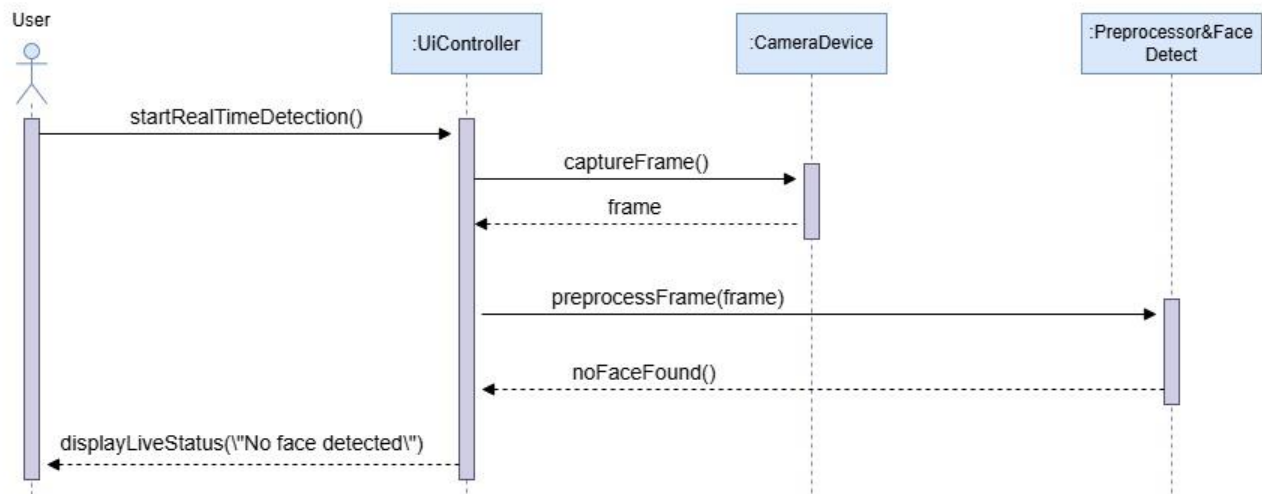
Upload Image for Face Mask Detection



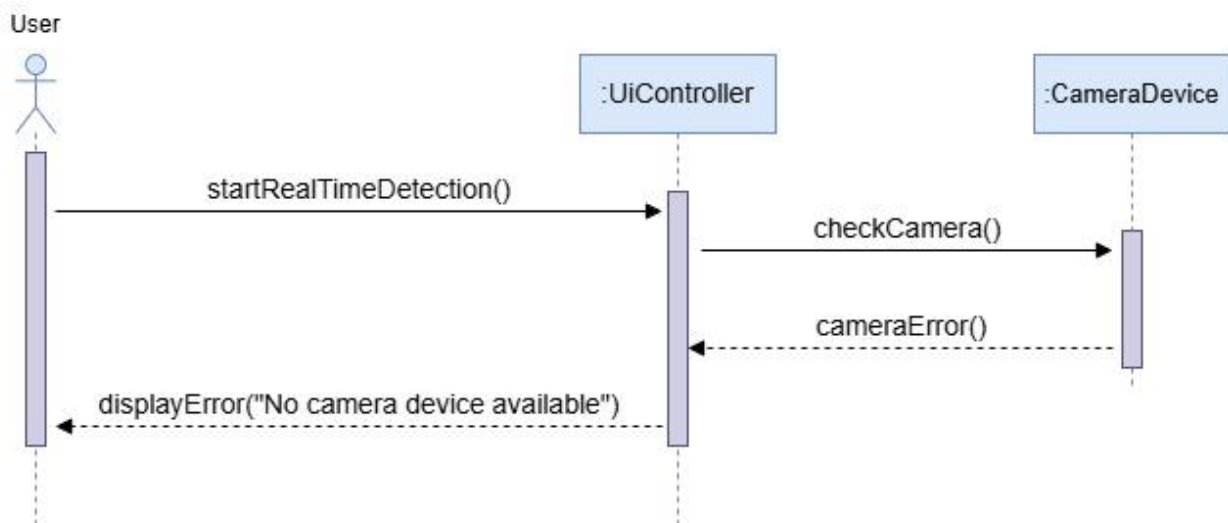
Real-Time Face Mask Detection (Webcam)



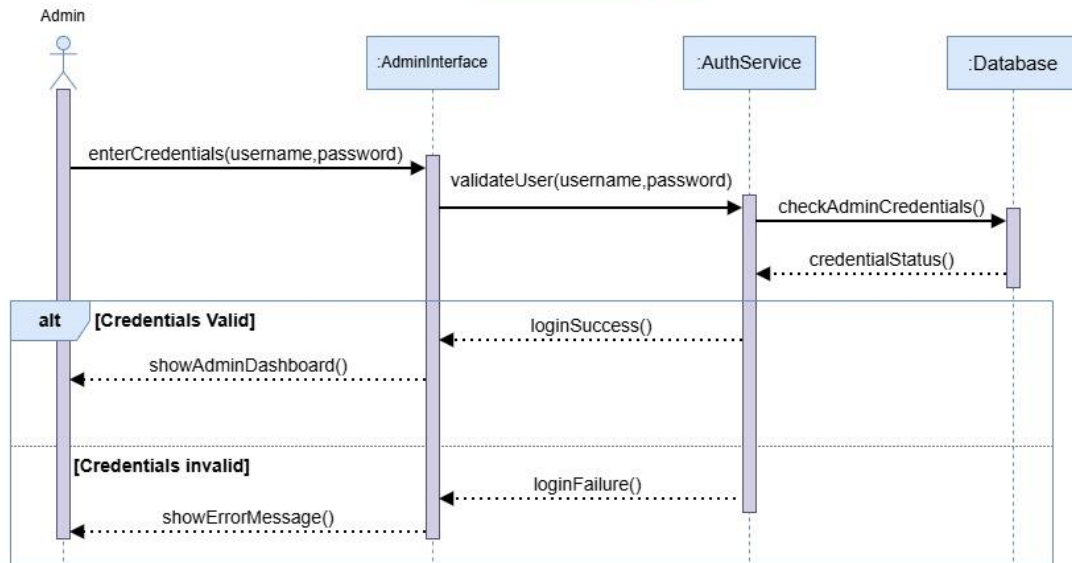
Face Not Detected in Real-Time Detection - (Exception)



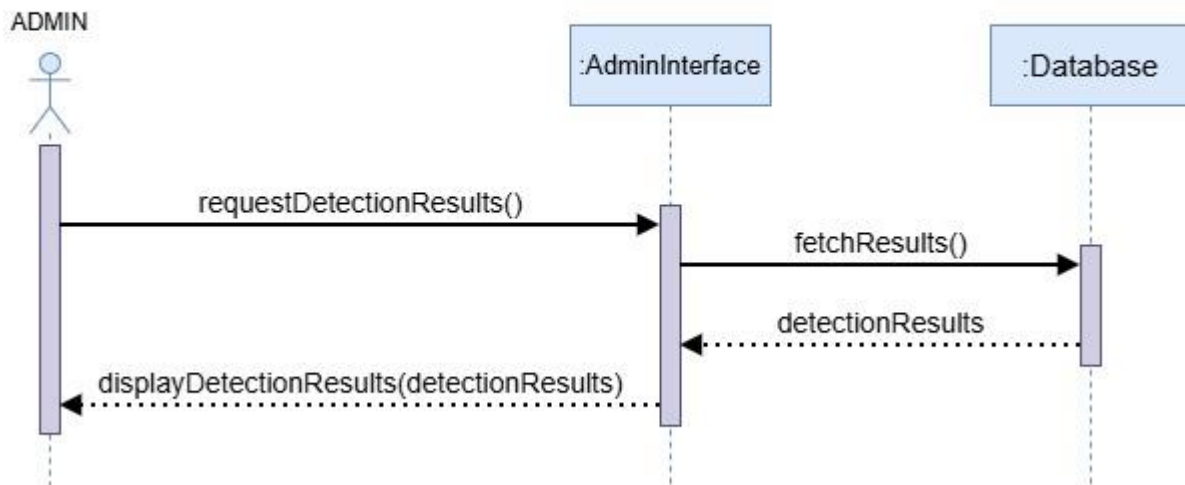
Camera Not Available Exception in real Time Detection



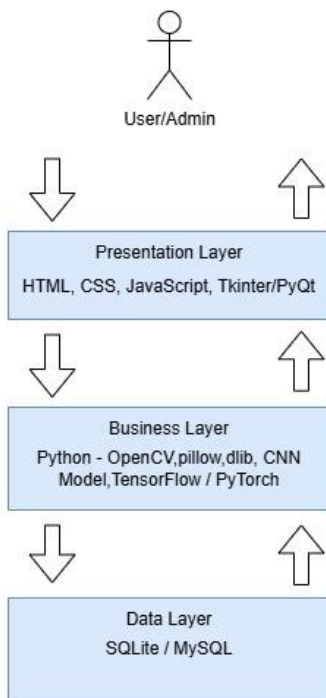
Admin Login



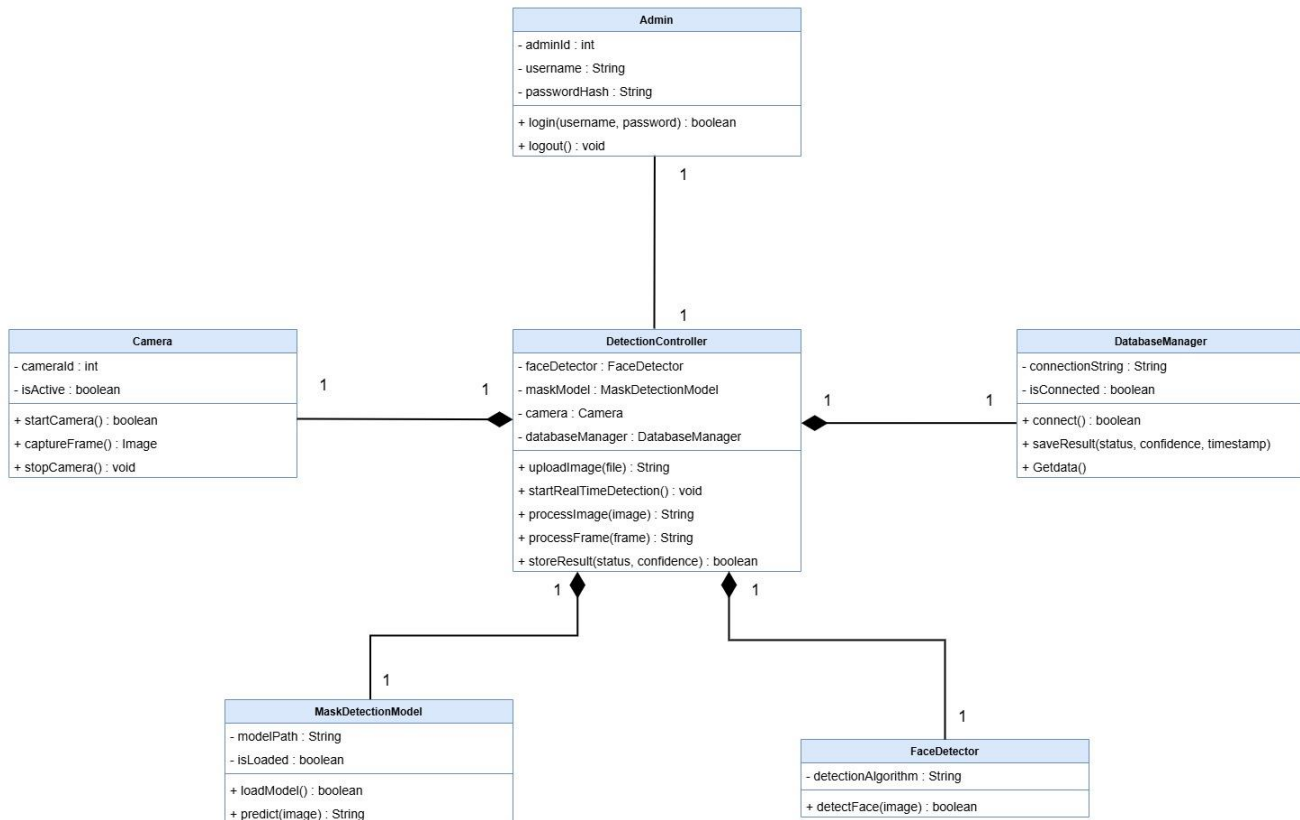
Admin View detection Results



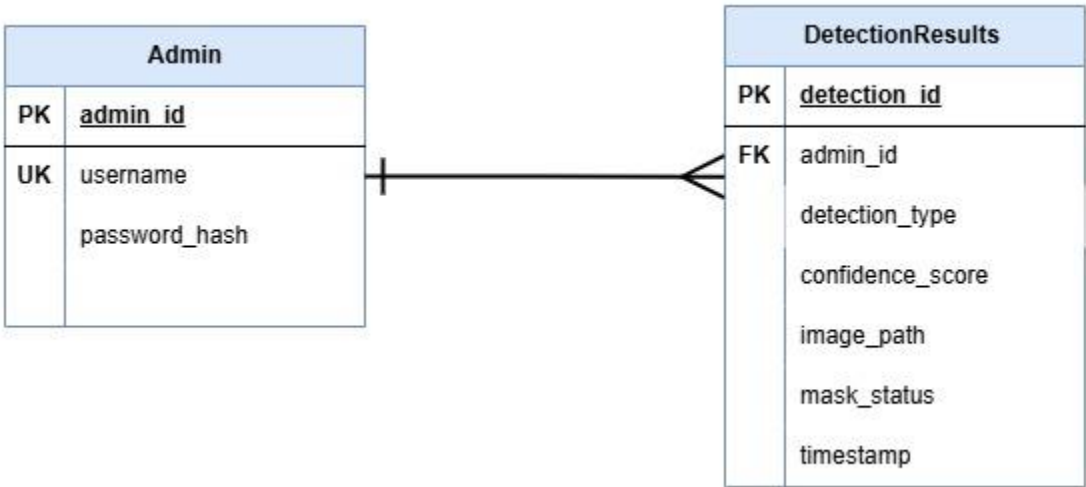
4. Architecture Design Diagram



5. Class Diagram



6. Database Design



7. Interface Design

Home Screen

**Face Mask Detection System**
AI-Powered Detection Platform

Admin Login

Welcome to Face Mask Detection System

Choose your preferred detection method to get started. Our system uses advanced AI to detect face mask status in real-time or from uploaded images.



Upload Image

Upload an image to detect face mask status

Get Started →



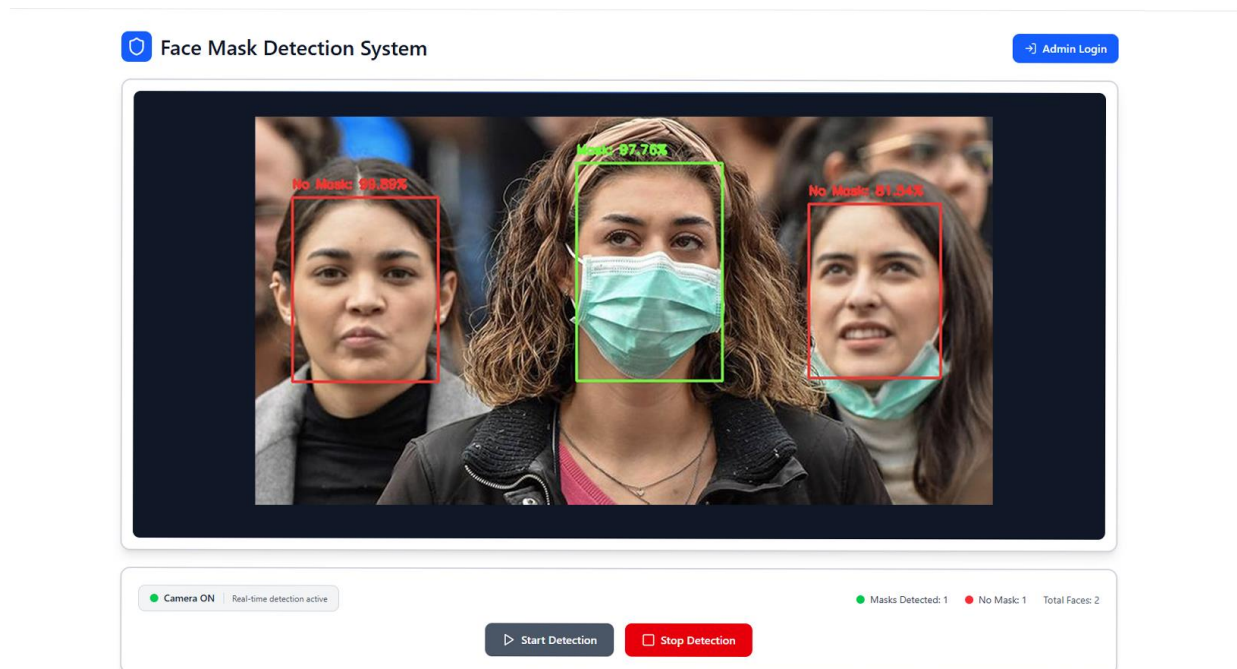
Real-Time Detection

Use live camera feed for real-time face mask detection

Get Started →

● System Ready

Real-Time Detection Screen



Admin Login



Face Mask Detection System

Admin Access Portal

Username

Password



Login

For testing: username: admin / password: admin123

© 2026 Face Mask Detection System. Academic Project.

Admin Dashboard

Admin Dashboard

Face Mask Detection System

Total Detections
1247



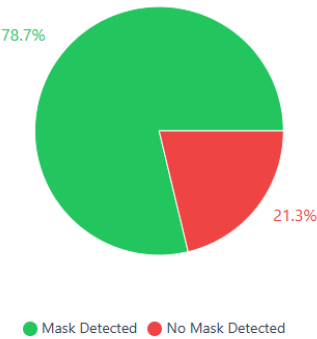
Mask Detected
982



No Mask Detected
265



Detection Distribution

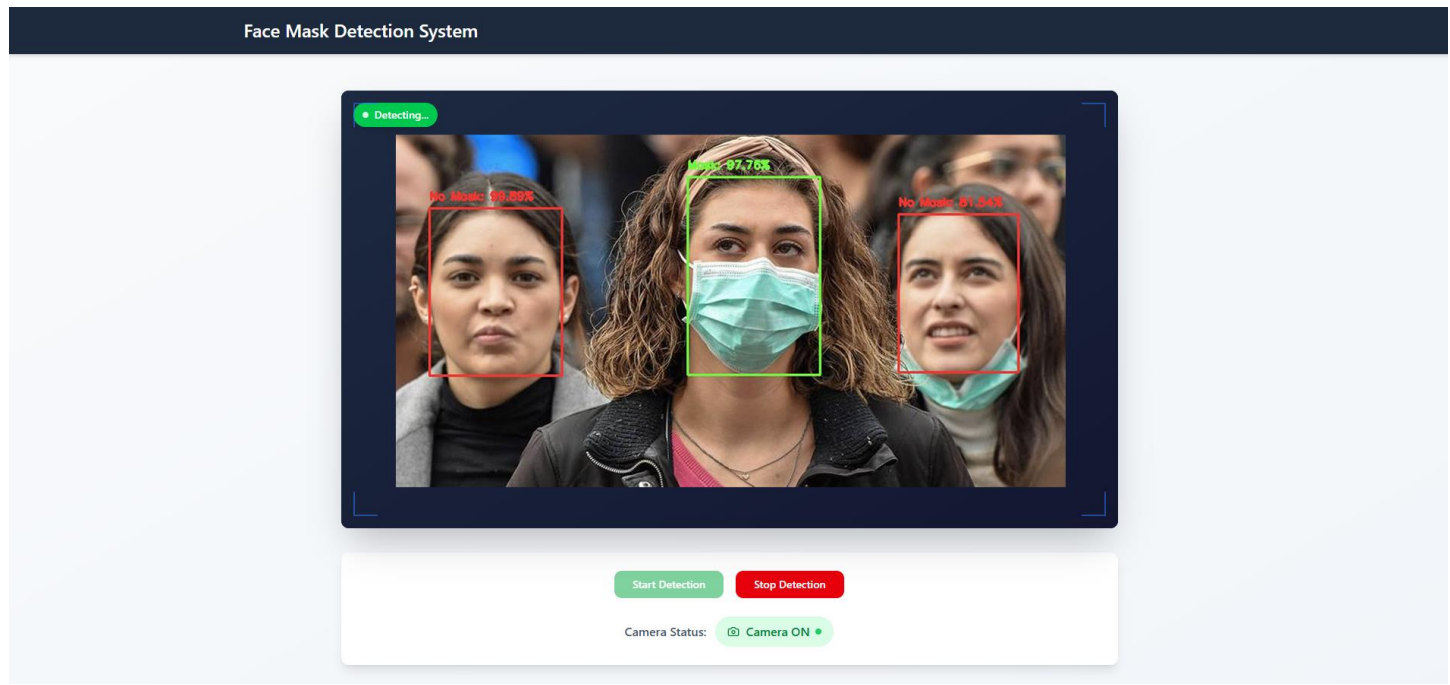


Navigation

[📄 Detection Screen](#)

[🚪 Logout](#)

Web Interface



8. Test Cases

TC-01 Upload Image for Detection

Field	Description
Test Case ID	TC-01
Test Case Description	Verify that a user can upload a valid image for mask detection.
Pre-Conditions	System is running; user is on the upload image screen.
Test Steps	1. Click "Upload Image".2. Select a valid JPG/PNG image (<5 MB).3. Submit.
Test Data	Valid JPG/PNG image with a clear face.
Expected Result	Image is accepted and forwarded for processing.
Post-Condition	System proceeds to "Preprocess & Detect Mask".
Actual Result	System accepted the uploaded image and forwarded it for processing in the "Preprocess & Detect Mask" module.
Status	Pass
Created By	BC220404931
Date of Creation	18 Jan 2026
Executed By	BC220404931
Date of Execution	18 Jan 2026

TC-02 Upload Invalid Image Format

Field	Description
Test Case ID	TC-02
Test Case Description	Verify that the system rejects invalid image formats.
Pre-Conditions	System is running; user is on the upload image screen.
Test Steps	1. Click "Upload Image".2. Select a non-image file (e.g., .txt, .pdf).3. Submit.
Test Data	Invalid file (e.g., document.pdf).
Expected Result	System shows error message: "Invalid format".
Post-Condition	User remains on the upload screen.
Actual Result	System rejected the uploaded file and displayed the error message “Invalid format,” keeping the user on the upload screen.
Status	Pass
Created By	BC220404931
Date of Creation	18 Jan 2026
Executed By	BC220404931
Date of Execution	18 Jan 2026

TC-03 Upload Image >5 MB

Field	Description
Test Case ID	TC-03
Test Case Description	Verify that system rejects images larger than 5 MB.
Pre-Conditions	System is running; user is on upload screen.
Test Steps	1. Click "Upload Image".2. Select image larger than 5 MB.3. Submit.
Test Data	Large image (>5 MB).
Expected Result	System shows error: "File size exceeds limit".
Post-Condition	User stays on upload screen.
Actual Result	System rejected the uploaded image and displayed the error message “File size exceeds limit,” keeping the user on the upload screen.
Status	Pass
Created By	BC220404931
Date of Creation	18 Jan 2026—
Executed By	BC220404931
Date of Execution	18 Jan 2026

TC-04 Start Real-Time Detection Successfully

Field	Description
Test Case ID	TC-04
Test Case Description	Verify real-time detection starts when camera is available.
Pre-Conditions	Camera connected; permissions granted.
Test Steps	1. Click "Start Real-Time Detection".2. Allow camera access if prompted.
Test Data	N/A
Expected Result	Webcam initializes; live feed starts; frames are sent for processing.
Post-Condition	System begins detecting masks in real-time.
Actual Result	Real-time detection started successfully; webcam initialized, live feed displayed, and frames were sent for mask detection processing
Status	Pass
Created By	BC220404931
Date of Creation	18 Jan 2026
Executed By	BC220404931
Date of Execution	18 Jan 2026

TC-05 No Camera Device Available

Field	Description
Test Case ID	TC-05
Test Case Description	Verify error when no camera is detected.
Pre-Conditions	No camera connected or camera disabled.
Test Steps	1. Click "Start Real-Time Detection".
Test Data	N/A
Expected Result	System shows error: "No camera device available".
Post-Condition	User remains on main screen.
Actual Result	System did not detect any camera device and displayed the error message "No camera device available," keeping the user on the main screen.
Status	Pass
Created By	BC220404931
Date of Creation	18 Jan 2026
Executed By	BC220404931
Date of Execution	18 Jan 2026

TC-06 Face Detected and Mask Status Determined

Field	Description
Test Case ID	TC-06
Test Case Description	Verify system correctly processes face and classifies mask status.
Pre-Conditions	Valid preprocessed image available.
Test Steps	1. System detects face in image.2. Resizes/normalizes image.3. CNN model predicts mask/no mask.
Test Data	Preprocessed face image.
Expected Result	System returns classification (Mask/No Mask) with confidence score.
Post-Condition	Result sent to UI and database.
Actual Result	System successfully detected the face, processed the image, and classified the mask status (Mask/No Mask) along with an accurate confidence score
Status	Pass
Created By	BC220404931
Date of Creation	18 Jan 2026
Executed By	BC220404931
Date of Execution	18 Jan 2026

TC-07 No Face Detected

Field	Description
Test Case ID	TC-07
Test Case Description	Verify system handles no-face scenario.
Pre-Conditions	Image uploaded or captured.
Test Steps	1. System runs face detection.2. No face is detected.
Test Data	Image without a face.
Expected Result	System displays error: "No face detected".
Post-Condition	User can retry with a new image.
Actual Result	System did not detect any face in the provided image and displayed the message “No face detected,” allowing the user to retry with a new image.
Status	Pass
Created By	BC220404931
Date of Creation	18 Jan 2026
Executed By	BC220404931
Date of Execution	18 Jan 2026

TC-08 CNN Model Fails to Load

Field	Description
Test Case ID	TC-08
Test Case Description	Verify error when CNN model fails to load.
Pre-Conditions	System attempts to load CNN model.
Test Steps	1. Initiate mask detection.2. System tries to load model.3. Model fails to load.
Test Data	N/A
Expected Result	System shows error: "Model failed to load".
Post-Condition	Detection halted.
Actual Result	System failed to load the CNN model and displayed the error message “Model failed to load,”
Status	Pass
Created By	BC220404931
Date of Creation	18 Jan 2026
Executed By	BC220404931
Date of Execution	18 Jan 2026

TC-09 Store Detection Result in DB Successfully

Field	Description
Test Case ID	TC-09
Test Case Description	Verify detection result is stored in database.
Pre-Conditions	Valid mask detection results generated.
Test Steps	1. System prepares data (status, timestamp).2. Connects to DB.3. Inserts record.
Test Data	Mask status, timestamp, confidence score.
Expected Result	Record saved successfully in database.
Post-Condition	Log entry created for viewing.
Actual Result	Detection result was successfully inserted into the database with correct mask status, timestamp, and confidence score.
Status	Pass
Created By	BC220404931
Date of Creation	18 Jan 2026
Executed By	BC220404931
Date of Execution	18 Jan 2026

TC-10 Database Connection Failure

Field	Description
Test Case ID	TC-10
Test Case Description	Verify error when DB connection fails.
Pre-Conditions	System tries to store detection result.
Test Steps	1. System attempts DB connection.2. Connection fails.
Test Data	N/A
Expected Result	System shows error: "Database connection failure".
Post-Condition	Record not saved; detection continues without storage.
Actual Result	System failed to establish a database connection and displayed the error message "Database connection failure,"
Status	Pass
Created By	BC220404931
Date of Creation	18 Jan 2026
Executed By	BC220404931
Date of Execution	18 Jan 2026

TC-11 Admin Login Successfully

Field	Description
Test Case ID	TC-11
Test Case Description	Verify admin can log in with valid credentials.
Pre-Conditions	Admin account exists.
Test Steps	1. Navigate to admin login.2. Enter valid username/password.3. Submit.
Test Data	Valid admin credentials.
Expected Result	Login successful; admin dashboard displayed.
Post-Condition	Admin can view detection results and stats.
Actual Result	Admin logged in successfully using valid credentials and the admin dashboard was displayed
Status	Pass
Created By	BC220404931
Date of Creation	18 Jan 2026
Executed By	BC220404931
Date of Execution	18 Jan 2026

TC-12 Admin Login with Wrong Credentials

Field	Description
Test Case ID	TC-12
Test Case Description	Verify system rejects invalid login.
Pre-Conditions	Admin login page is open.
Test Steps	1. Enter invalid username/password.2. Submit.
Test Data	Invalid credentials (e.g., wrong password).
Expected Result	Error message: "Wrong credentials".
Post-Condition	User stays on login page.
Actual Result	System rejected the login attempt and displayed the error message "Wrong credentials."
Status	Pass
Created By	BC220404931
Date of Creation	18 Jan 2026
Executed By	BC220404931
Date of Execution	18 Jan 2026

TC-13 View Detection Logs & Stats

Field	Description
Test Case ID	TC-13
Test Case Description	Verify admin can view detection results.
Pre-Conditions	Admin is logged in; detection result exist in DB.
Test Steps	1. Navigate to detection results section.2. System retrieves and displays detection results
Test Data	N/A
Expected Result	Logs and statistics displayed correctly.
Post-Condition	Admin can analyze historical data.
Actual Result	Logs were successfully retrieved from the database and displayed correctly with relevant statistics.
Status	Pass
Created By	BC220404931
Date of Creation	18 Jan 2026
Executed By	BC220404931
Date of Execution	18 Jan 2026